

Original Article

Craniofacial and Axial Skeletal Defects Induced by the Fungicide Triadimefon in the Mouse

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BACKGROUND: Triadimefon is an antifungal derived from triazole. In in vitro whole-rodent embryo cultures, triazole-derivatives showed specific teratogenic effects at the branchial apparatus. The aim of the present work was to test in vivo triadimefon (FON), in order to verify a relationship between triazole exposure, embryonic abnormalities, and/or fetal malformations. **METHODS:** Pregnant CD-1 mice were treated with 0–300 mg/kg FON by gavage on day 8 post coitum (p.c.) at 10:00 AM, and sacrificed on day 8 p.c. at 1:00 PM, on day 9 p.c. at 10:00 AM, on day 10 p.c. at 10:00 AM, and at term of gestation (day 18 p.c.). At midgestation, the embryos were processed for specific immunostainings to visualize the hindbrain segmentation (day 8 p.c.) and the neural crest cell migration (days 8 and 9 p.c.). Fetuses explanted at term were all processed for skeletal examination after double-staining of osseous and cartilaginous tissues. **RESULTS:** At midgestation, the immunostaining of rhombomeres 3 and 5 showed a light scattering of the immunostained areas; the neural crest cell migration was unaffected, but their localization at the branchial arch level was abnormal. At term, several severe malformations were observed at the craniofacial and at the axial skeletal level. Ectopic cartilage was observed at the upper jaw. **CONCLUSIONS:** Triadimefon is teratogenic. The observed craniofacial malformations could be explained by an alteration of the rhombomeric organization and neural crest migration to the branchial arches; the axial abnormalities could be explained by the abnormal segmental identity specification. *Birth Defects Res B* 74:185–195, 2005. © 2005 Wiley-Liss, Inc.

Key words: triazole; teratogenesis; homeotic respecification; birth defects

INTRODUCTION

Triazole-derivatives constitute a class of substances containing one or two triazole groups in their molecular formula. Classified as antifungals, they are widely used as agrochemicals against mildews and rusts of cereal grains, fruits, vegetables, and ornamentals, or clinically against topical or disseminated candidiasis and coccidioid meningitis. These compounds inhibit fungal proliferation by their interference with steroid biosynthesis and fungal cell-wall formation mediated by P450 enzymes (Vanden Bossche et al., 1984).

Conventional teratological in vivo studies showed that these substances are slightly teratogenic after daily oral administration during the whole embryogenic period both in rats and in rabbits. Teratogenic effects, in fact, occur only at maternotoxic dose levels. Interestingly, however, the observed malformations are common to all tested triazoles (agrochemicals: triadimefon [FON], triadimenol, flusilazole, bitertanol, cyproconazole; clinical drugs: fluconazole, itraconazole), and are described as cleft palate (Machemer, 1976, 1977; Korte and Ostenburg, 1980; FAO, 1982, 1985a, 1985b, 1989; Unger et al., 1982; Becker et al., 1986; Tachibana et al., 1987; Van Cauteren et al., 1987; Vergieva, 1990; Machera, 1995).

Even if no data are available on the effect of agrochemical triazoles on human pregnancy, the effect of the clinical drug fluconazole on newborns has been documented, with contrasting results. In fact, while prospective studies showed that the single use of 50–150 mg/kg of fluconazole during the first trimester of pregnancy is not related to an increased teratological risk, case-reports, referring to long-term or high-dose (400–800 mg/day) maternal fluconazole therapy, reported craniofacial and limb abnormalities in newborns (Lee et al., 1992; Mastroiacovo et al., 1996; Pursley et al., 1996; Wiesinger et al., 1996; Aleck and Bartley, 1997; Sanchez and Moya, 1998).

Recently, postimplantation rodent embryos cultured in vitro from the embryonic stage corresponding to 1–3 somites (rat) or 8–10 somites (mouse) for 48 hours in presence of some triazoles (triadimefon, triadimenol, flusilazole, and fluconazole for rats; fluconazole for mice)

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showed severe and specific alterations at the branchial apparatus level (reduction and fusion of the first and second branchial arches, and agenesis of the second branchial arch) (Tiboni, 1993; Menegola et al., 2000, 2001a), related to abnormal rhombencephalic organization (rhombencephalic segments 3 and 5 reduced and scattered after immunostaining) (Menegola et al., 2003, 2004), to abnormal neural crest cell migration from the rhombencephalon to the branchial arches (after immunostaining, neural crest cells converged in a unique flow instead of forming three separate migrating streams as observed in controls) (Menegola et al., 2003, 2004), and to the formation of abnormal ectomesenchymal tissue from migrated neural crest cells (after 48 hours of culture, the neural crest cells reached the branchial apparatus but were unable to condense) (Menegola et al., 2003, 2004). In mammalian embryos, the pharyngeal ectomesenchyme, originated by the cranial neural crest cells that migrated into the branchial arches, gives rise to the craniofacial bones and to the ossicles of the middle ear (Johnston et al., 1985) (Fig. 1).

The aim of the present work was to study, using an unconventional *in vivo* protocol (single oral administration instead of daily treatment during the whole embryogenic period), the teratogenic potential in mice of one triazole derivative, FON, as representative of the whole chemical family. To compare the results with data previously obtained *in vitro*, the treatment time (day 8 post coitum [p.c.], 10:00 AM) was chosen considering the embryonic stage (5–13 somites, as established by a previous staging test), and dams were sacrificed at midgestation (at times corresponding to the embryonic stages comparable to those analyzed during the *in vitro* protocols) or at term.

As branchial arch ectomesenchyme produces exquisitely facial skeletal structures, fetuses explanted at term were all processed for skeletal analysis.

MATERIALS AND METHODS

Virgin CD-1 mice (Charles-River, Calco, Italy), housed in a thermostatically maintained room ($T = 22 \pm 2^\circ\text{C}$;

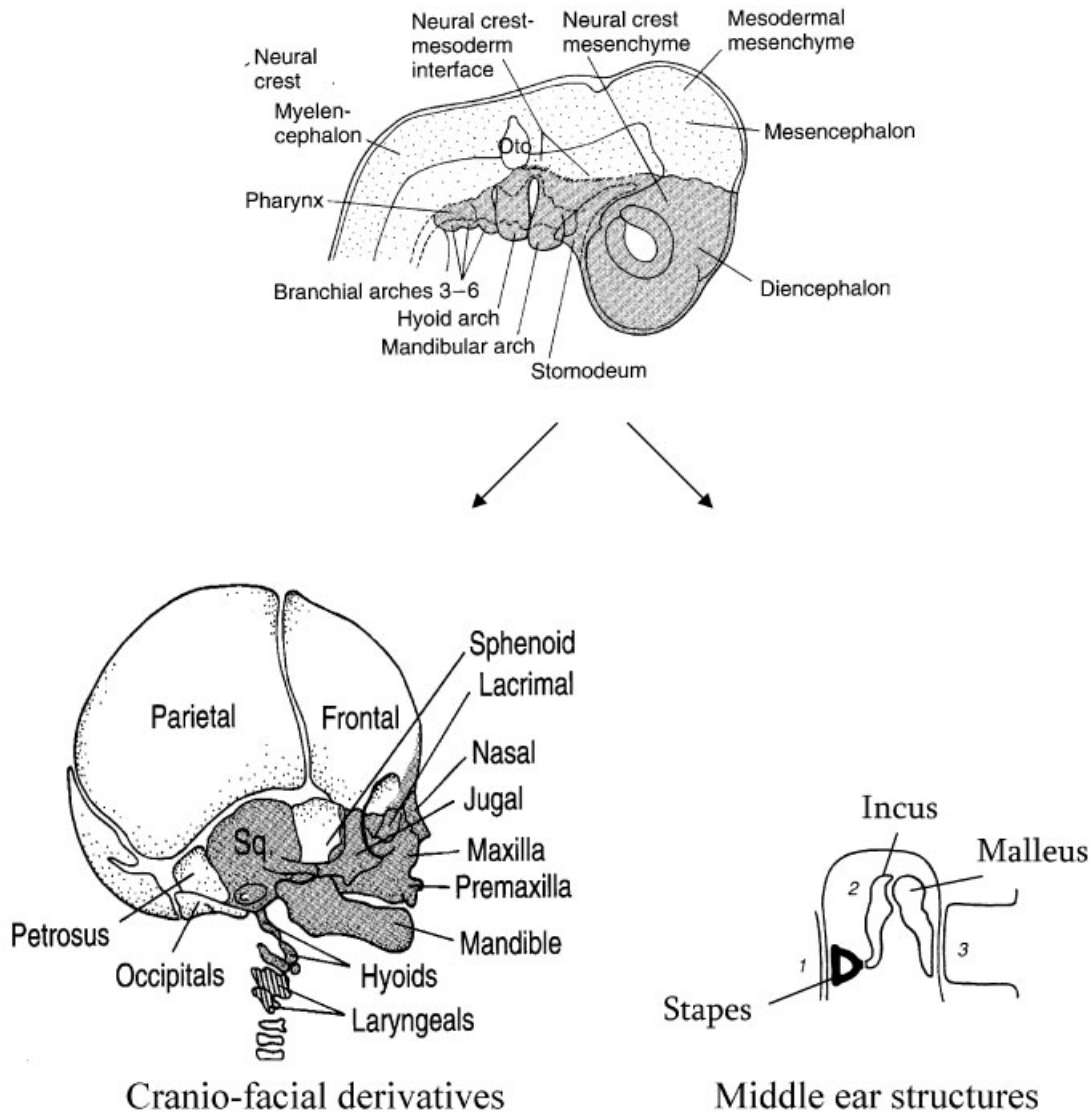


Fig. 1. Scheme showing the skeletal structures originating from the first and the second branchial arches. Sq: squamosal.

relative humidity $55 \pm 5\%$; light cycle from 7:00 AM to 7:00 PM) with food (Italiana Mangimi, Italy) and tap water ad libitum, were caged overnight with males of proven fertility. The day of the vaginal plug was considered as GD 0. Pregnant mice were orally treated by gavage with 0 or 300 mg/kg FON (gift of Bayer, Italy; dissolved in a solution containing 10% acetone in corn oil) on day 8 p.c. at 10:00 AM. The treatment time was selected on the basis of a previous staging test showing that at this time the embryonic stage ranges from 5 to 13 somites. The FON dose level was selected on the basis of a previous range finding test, showing 300 mg/kg as the highest dose well tolerated by this strain of mice. This unconventional protocol was chosen to compare the results obtained in *in vitro* postimplantation whole embryo culture with an *in vivo* exposure of embryos at similar developmental stages. As it was not a proposal of this study to individuate a no observed effect level (NOEL) for risk assessment, but just to demonstrate the intrinsic teratogenic properties of FON as representative of a class of triazoles, the highest well tolerated dose level was selected, on the basis of a previous range finding test on FON 300–400–500 mg/kg.

Analysis at Midgestation

Females were sacrificed by CO₂ asphyxia at midgestation in order to evaluate their rhombencephalic organization, neural crest cell migration into the branchial arches, and branchial arch ectomesenchymal compaction.

Females were sacrificed on day 8 p.c. at 1:00 PM to evaluate by immunostaining rhombomeres 3, 4, and 5, as described by White et al. (2000). For this purpose, only embryos at the 8–13 somite stage were chosen for staining. The primary antibodies used were an anti-Krox 20 protein antibody to mark rhombomeres 3 and 5 and an anti-Hox-b1 protein antibody to mark rhombomere 4. The procedures of whole mount immunostaining are described in Table 1.

To evaluate neural crest cell migration and ectomesenchyme compaction, neural crest cells were immunolocalized according to the method described by Wei et al. (1999) by the use of an anticellular retinoic acid binding protein (anti-CRABP) antibody on embryos explanted on day 8 p.c. at 1:00 PM (embryonic stage 8–18 somites) and

on day 9 p.c. at 10:00 AM (embryonic stage 23–26 somites). The procedures of whole-mount immunostaining are described in Table 1. After staining, embryos were analyzed under a dissecting microscope and photographed.

Analysis at Term

After dam sacrifice by CO₂ asphyxia, the entire uterine content was weighed and total implantation, live and dead fetuses, early and late resorptions were recorded. The postimplantation loss index [(total implantation – live fetuses/total implantation) $\times$ 100] was calculated. Live fetuses were then explanted from the uterus, morphologically examined, and weighed. All the explanted fetuses were processed for the double skeletal staining of bone and cartilage, following the method described by Kimmel and Trammell (1981), partially modified as previously described (Menegola et al., 2001b). The osseous and cartilaginous tissues were stained using respectively alizarin red and alcian blue reagents. After staining, fetuses were analyzed under a dissecting microscope and the abnormalities were recorded.

Statistical Analysis

Statistical analysis was performed with the Student's *t*-test and the chi-square test. The level of significance was set at $p < 0.05$.

RESULTS

Analysis at Midgestation

A total of 35 embryos from controls (5 embryos/dam) and of 55 embryos (5 embryos/dam) from females treated with 300 mg/kg FON, all with a stage ranging from 8 to 13 somites, were explanted at day 8 p.c. (1:00 PM) and immunostained with anti-Krox 20 protein antibody to visualize rhombomeres 3 and 5 (r3, r5). In control embryos, immunostaining revealed two dark stripes with well-defined borders. In embryos explanted from FON-treated dams, the immunostained regions appeared reduced in height and scattered (Fig. 2a and b).

Table 1
Immunostaining Procedures

Stained structures	Fixation	Primary antibody	Secondary antibody	Visualization
Rhombomere 4	Dent's fixative (DMSO:-methanol 1:4)	Rabbit anti-Hox-b1 protein (Eurogentec):PBS 1:250	Goat anti-rabbit -IgG-peroxidase (Boehringer):PBS 1:40	4-Cl-1-naphtol (Sigma) 0.25% in ethanol 40% + oxygen peroxide 0.006%
Rhombomeres 3, 5	Overnight -20°C	Three days 4°C	Overnight 4°C	4-Cl-1-naphtol (Sigma) 0.25% in ethanol 40% + oxygen peroxide 0.006%
	Dent's fixative (DMSO:-methanol 1:4)	Rabbit anti-Krox-20 protein (Eurogentec):PBS 1:250	Goat anti-rabbit -IgG-peroxidase (Boehringer):PBS 1:40	
Neural crest cells	Overnight -20°C	Three days 4°C	Overnight 4°C	4-Cl-1-naphtol (Sigma) 0.25% in ethanol 40% + oxygen peroxide 0.006%
	Dent's fixative (DMSO:-methanol 1:4)	Mouse anti-CRABP (ABR):PBS 1:250	Goat anti-mouse -IgG-peroxidase (Boehringer):PBS 1:40	
	Overnight -20°C	Three days 4°C	Overnight 4°C	

DMSO = dimethylsulfide; PBS = phosphate saline buffer; IgG = immunoglobulin G.

Eurogentec, Covance Barkeley, Richmond, CA; Boehringer, Mannheim, Germany; Sigma, St. Louis, MO; ABR, Golden, CO.

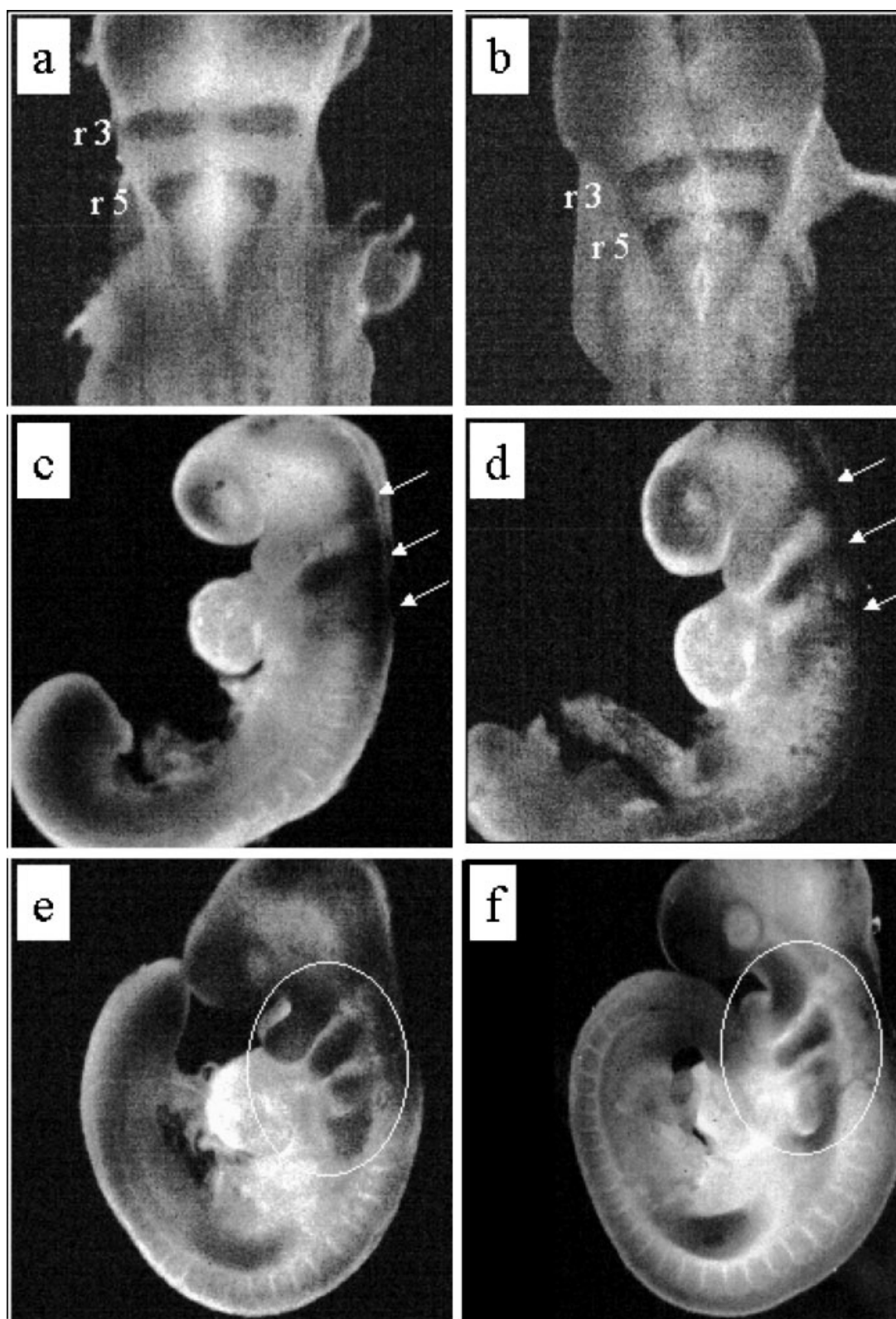


Fig. 2. a,b: Dorsal view of the hindbrain of day 10 p.c. embryos immunostained with the antibody anti-Krox-20 protein showing the structure of rhombomeres 3 and 5. (20 $\times$) a: Control. b: Embryo explanted from a FON 300 mg/kg treated dam. Note that the immunostained regions appear light scattered and reduced in height. c-f: Lateral view of day 10 p.c. (15 $\times$; c,d) and day 11 p.c. (12 $\times$; e,f) embryos immunostained with an anti-CRABP antibody. Three separate flow (arrows) of migrating neural crest cells are visible from the hindbrain (from the regions corresponding to rhombomeres 2, 4, and 6) to the branchial region (to the morphogenic region of the first, second, and posterior branchial arches, respectively) both in the control (c) and in an embryos explanted from a FON 300 mg/kg treated dam (d). After migration, however, the ectomesenchyme derived from the migrated neural crest cells (dark tissue in the circled area) appears abnormally distributed in embryos explanted from the FON group dams (f) in comparison to controls (e).

When immunostained with the antibody anti-Hox-b1 protein, on the contrary, the 35 control and 55 FON embryos, explanted from the same litters used for Krox-20 analysis, showed a similar pattern: a unique, compact, dark stripe with well-defined borders.

To visualize neural crest cell migration from the rhombencephalon to the branchial arches, 35 and 55 embryos explanted from seven controls and from 11 FON-treated dams at day 8 p.c. (1:00 PM), respectively, and 25 and 25 embryos explanted from five controls and from 5 FON-treated dams at day 9 p.c. (10:00 AM), respectively, were immunostained with antibody anti-CRABP. In the younger embryos (8–18 somites), both in controls and in FON-exposed conceptuses, three very distinct immunostained cell flows, directed from the rhombencephalon to the branchial arches, were observed. No apparent differences were observed between the two groups (Fig. 2c and d). Control embryos with a stage ranging from 23 to 26 somites showed three distinct and very condensed ectomesenchymal areas at the first and the second branchial arches and at the region originating at the more caudal branchial arches. Embryos of the same developmental stage explanted from FON-treated dams showed apparently normal-shaped branchial arches. However, CRABP immunostaining revealed a clear abnormal compaction of neural crest cells, which had migrated into the branchial apparatus (Fig. 2e and f).

Analysis at Term

As shown in Table 2, the postimplantation loss index and the mean fetal weight were similar to those of the control group.

Table 2
Litter Parameters at Term of Pregnancy

	Control	FON 300 mg/kg
Mated females	7	12
Females with implantations	5	9
Females with live fetuses	5	9
Implantations		
Total	62	129
M ± SD	12.40 ± 1.52	14.33 ± 2.12
Live fetuses		
Total	57	125
M ± SD	11.40 ± 1.95	13.88 ± 2.42
Postimplantation loss (%)	8.06	2.63
Fetal weight (M ± SD)	1.42 ± 0.05	1.33 ± 0.07

Table 3
External Abnormalities (%)

	CON	FON 300 mg/kg
Total litters	5	9
Total examined fetuses	54	121
Fetuses with external defects	0	25.62 ^a
Cleft palate	–	23.14
Micrognathia	–	1.65
Macrognathia	–	0.83

^a*p* < 0.01 vs. Control.

Table 4
Skeletal Abnormalities Detected at the Craniofacial Level (%)

	CON	FON 300 mg/kg
Total litters	5	9
Total examined fetuses	54	121
Fetuses with craniofacial defects	9.3	86.8 ^a
Fusions		
Ear ossicles		47.1
Tympanic ring-ear ossicles		23.1
Squamosal-tympanic ring		9.1
Squamosal-zygomatic		6.6
Tympanic ring-alisphenoid		5.0
Tympanic ring-ear ossicles-Meckel's cartilage		4.2
Abnormal shape		
Squamosal-Meckel's cartilage		3.3
Tympanic ring-ptyergoid		3.3
Squamosal-ptyergoid		2.5
Squamosal-styloid process		1.7
Squamosal-tympanic ring-ptyergoid		1.7
Tympanic ring-Meckel's cartilage		1.7
Squamosal-alisphenoid		1.7
Basioccipital-basisphenoid		0.8
Squamosal-ear ossicles		0.8
Squamosal-alisphenoid-ptyergoid		0.8
Tympanic ring-alisphenoid-ptyergoid		0.8
Abnormal shape		
Tympanic ring		81.8
Squamosal		52.8
Zygomatic		42.1
Styloid process		29.0
Alisphenoid		14.9
Pterygoid		9.9
Mandible		2.5
Palatine		0.8
Agnesis		
Styloid process		10.7
Rostral process of the Meckel's cartilage	9.3	6.6
Tympanic ring		0.8
Ectopic cartilage		
Upper jaw		76.9

^a*p* < 0.01 vs. Control.

A total of 26% of FON-exposed fetuses showed external malformations (Table 3). In particular, cleft palate (28 cases), macrognathia (1 case), and micrognathia (2 cases) were observed. No abnormal fetuses were observed from controls.

As far as the skeletal analysis is concerned, after double-staining for bone and cartilage, 9% of control fetuses and 87% of FON-exposed fetuses showed craniofacial defects (Table 4). The observed malformations were all at the level of skeletal structures derived from the first and second branchial arches (Fig. 1; Table 4) and were: fusion, agnesis, and abnormal shape of osseous or cartilaginous elements of the jaw, of the basal and lateral cranial structures, and of the middle ear ossicles (Fig. 3).

Interestingly, axial defects were recorded in 87% of fetuses explanted from FON-treated dams (Table 5). The abnormalities were fusions, duplications, or morphological transformations of axial segments. The analysis of phenotypes revealed that all controls showed the typical axial formula with 7 cervical, 13 thoracic (with ribs, of

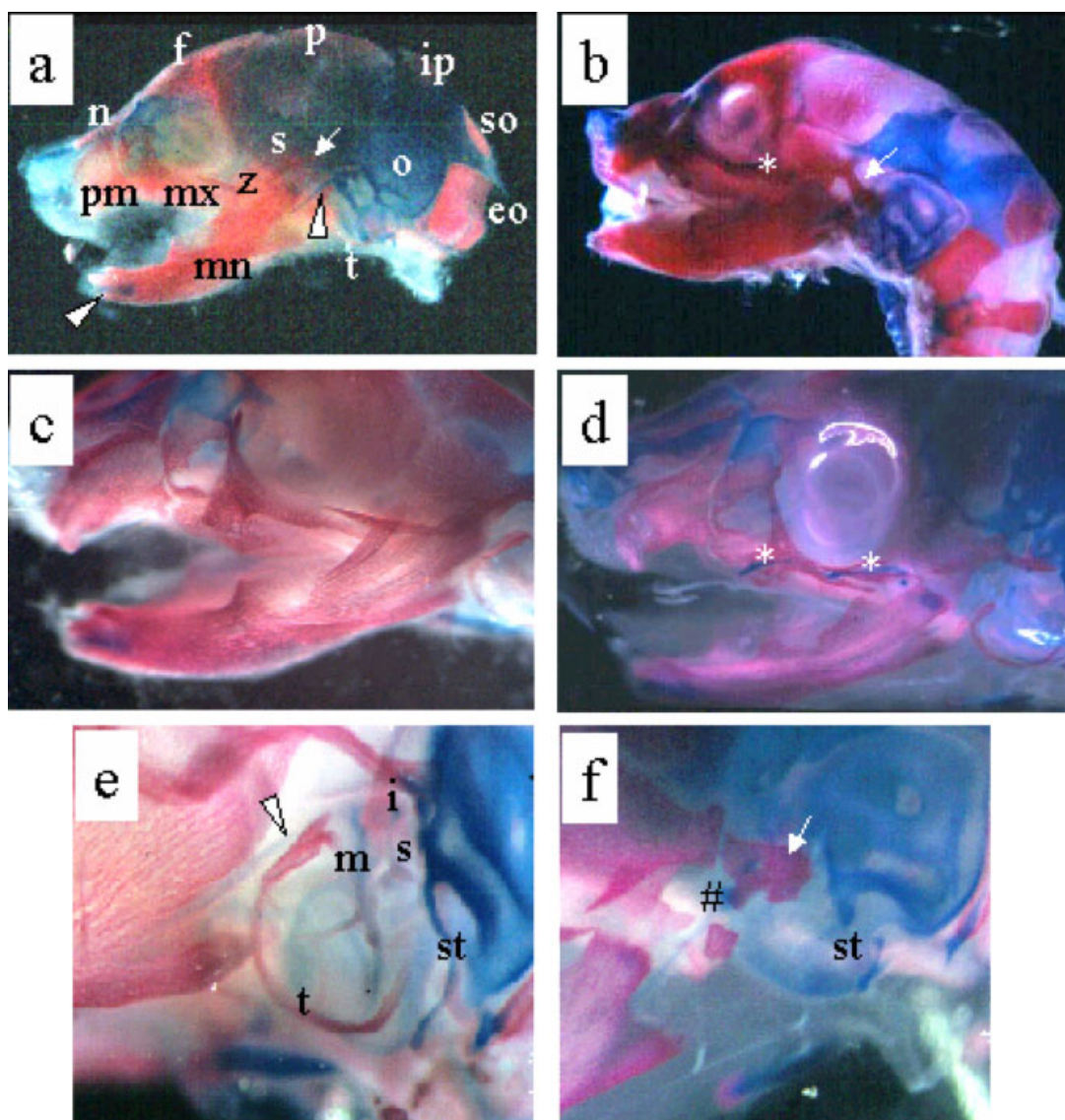


Fig. 3. Lateral view of the skull of fetuses explanted from a controls (a,c,e) and from FON 300 mg/kg treated dams (b,d,f), stained using the double staining method for bone (red) and cartilage (blue). a,b: $8\times$; c,d: $20\times$; e,f: $25\times$. The following structures are visible: n, nasal; f, frontal; p, parietal; ip, interparietal; so, supraoccipital; eo, exoccipital; pm, premaxilla; mx, maxilla; z, zygomatic; s, squamosal (the arrows show the retrotympic process); mn, mandible; o, otic capsule; t, tympanic ring; and st, styloid process. Arrowheads show the rostral and distal processes of the Meckel's cartilage. Note in b and f the abnormal shape of the retrotympic process, and in b and d the ectopic cartilage visible at the upper jaw (*) in FON 300 group explanted fetuses. In e, note the normal structures of the ear ossicles malleus (m), incus (i), and stapes (s) and of the distal region of the Meckel's cartilage (arrowhead), completely fused together in f in the FON 300 group fetus (#), where the tympanic ring agenesis and the abnormal and reduced styloid process is also visible.

those only seven reached the sternum in 86% of the cases, eight reached the sternum in 14% of the cases), 6 lumbar, 5 sacral, and a number of caudal vertebrae (Figs. 4a–c, 5a–c). In 40% of fetuses explanted from FON-treated dams, the phenotype was compatible with an anterior homeotic transformation (eight ribs reaching the sternum associated with a lumbar rib and in some cases to a lumbar morphology of the first and second sacral vertebrae) (Fig. 4d–f). In addition, 28% of the fetuses showed duplication of thoracic segments and, as a consequence, the axial formula was: 7 cervical, 14 thoracic, 6 lumbar, 5 sacral, and a number of caudal (Fig. 5d–f).

Ectopic cartilage at the level of the upper jaw, never recorded in controls, was observed in 77% of FON-exposed fetuses (Fig. 3b and d).

DISCUSSION

Single oral administration of 300 mg/kg FON on day 8 p.c. (10:00 AM, embryonic stage 5–13 somites) was highly teratogenic in mice. Fetuses explanted at term from FON-treated dams were, in fact, characterized by external (26% cleft palate) as well as skeletal (96%, craniofacial and axial, observed after double-staining for bone and cartilage) malformations. Previously published data on

Table 5
Skeletal Abnormalities Detected at the Axial Level (%)

	CON	FON 300 mg/kg
Total litters	5	9
Total examined fetuses	54	121
Fetuses with axial defects	35.0	87.0 ^a
Homeotic transformation of thoracic segments+lumbar rib	9.3	43.0
Cervical rib	16.7	21.5
Asymmetric sternum	5.6	19.0
Duplication of thoracic segments		17.3
Lumbar rib	31.5	11.6
8° rib reaching the sternum	3.7	6.6
Homeotic transformation of thoracic segments		5.8
6° lumbar vertebra morphologically transformed as sacral		5.8
8° rib near the sternum	1.9	4.1
1° sacral vertebra morphologically transformed as lumbar		4.1
2° sacral vertebra morphologically transformed as the 1°		4.1
Extra emi-sternebra without duplication		2.5
Fusions between ribs		1.7
Fusions between cervical vertebrae		0.8
7° rib far from the sternum		0.8

^a*p* < 0.01 vs. Control.

rat and rabbit teratogenesis of a number of triazole derivatives, administered daily during the whole embryogenic period, showed cleft palate as the typical malformation but, as it significantly occurred only at maternotoxic dose levels, it was related to maternal imbalance instead of a direct chemical effect (Machemer, 1976, 1977; Korte and Osteburg, 1980; FAO, 1982, 1985a, 1985b, 1989; Unger et al., 1982; Becker et al., 1986; Van Cauteren et al., 1987; Tachibana et al., 1987; Vergieva, 1990; Machera, 1995). For this reason, nowadays, both clinical and agrochemical triazole derivatives are commonly used against maternal infections and by pregnant workers.

The data obtained from this present work, using an unconventional nonmaternotoxic protocol, strictly suggest that the effects of the triazole FON on mice are not mediated by maternotoxicity. The aim of the present work was to determine the teratogenic potential of FON, as representative of the triazole family, and to relate the abnormalities elicited in vivo with the embryonic alterations previously described after in vitro exposure of postimplantation rat and mouse embryos to a number of triazoles (FON, triadimenol, flusilazole, fluconazole) (Tiboni, 1993; Menegola et al., 2000, 2001a, 2003, 2004). For this reason, some control and FON-treated females were sacrificed at midgestation and embryos were processed for immunolocalization of structures considered targets after in vitro analysis. Abnormal rhombencephalic organization at the level of rhombomeres 3 and 5, abnormal neural crest cell migration from hindbrain to the branchial arches, abnormal branchial arch ectomesenchyme compaction, and abnormal branchial arch morphology have been described after rat embryo in vitro exposure to fluconazole 500 μ M (Menegola et al.,

2003, 2004) or to FON 250 μ M, triadimenol 125 μ M, or flusilazole 25 μ M (Di Renzo et al., 2003). In the present study, embryos explanted at midgestation from FON-treated dams showed less severe but similar alterations at the level of the same target structures. In particular, abnormal organization of rhombomeres 3 and 5 (reduced in height and scattered, as they appeared after immunostaining of the characterizing protein Krox 20) and abnormal ectomesenchymal compaction at the level of the branchial arches (as observed after CRABP immunostaining) were detected. The correspondence between in vivo and in vitro results suggests that the in vitro model, at least for this class of substances, is adequate to determine the pathogenic pathway and predictive of teratogenic effects.

On the basis of the in vitro effects of triazoles on branchial arch morphology, craniofacial alterations were expected after in vivo exposure. Craniofacial morphogenesis is a complex phenomenon based on the interaction between different structures derived from neurocranial and branchial elements. The craniofacial and middle ear bones derived from the first and second branchial arch ectomesenchyme are summarized in Figure 1. As expected, a large number of the fetuses explanted at term from FON-treated dams were abnormal, after double-staining for bone and cartilage, specifically at the level of the craniofacial structures originating from the branchial ectomesenchyme (palatine, basisphenoid, alisphenoid, pterygoid, squamosal, zygomatic, maxilla, mandible, Meckel's cartilage, tympanic ring, and ear ossicles). The observed malformations were: abnormal shape, fusions, and agenesis. In addition to these abnormalities, ectopic cartilage was frequently observed at the level of the upper jaw. In primitive vertebrates, the upper jaw consists of the endoskeletal palatoquadrate cartilage originating from a part of the first branchial arch. The palatoquadrate, fully functional as jaw only in sharks, changes its functions in other vertebrates and, in mammals, becomes the incus of the middle ear and the alisphenoid bone, while the upper jaw is constituted by the dermal elements of premaxilla and maxilla. Craniofacial abnormalities were reported in the literature after exposure to different chemicals (Kay, 1987; Sulik et al., 1988; Rogers et al., 2004). On the contrary, ectopic upper jaw cartilage was only described on mouse embryos exposed to teratogenic dosages of retinoic acid (Morriss-Kay, 1993), a known morphogen acting during a number of embryogenic processes (gastrulation, hindbrain segmentation, axial segmentation, limb organization, and heart and great vessel formation), switching on the expression of genes of the Hox family (Morriss-Kay et al., 1991; Gudas, 1994; Clagett-Dame and Plum, 1997). After retinoic acid exposure, the ectopic cartilage was described as a duplication of the Meckel's cartilage (the second cartilaginous element derived from the first branchial arch, implied in the formation of the lower jaw in primitive vertebrates, encased in the dermal bone dentary in mammals). In fetuses exposed to retinoic acid, the duplicated Meckel's cartilage was completely displaced from the mandibular to the maxillary region (Morriss-Kay, 1993).

Unexpectedly, 87% of the fetuses explanted from FON-treated females also showed axial skeletal abnormalities, classified as fusions of vertebrae or ribs, duplications of

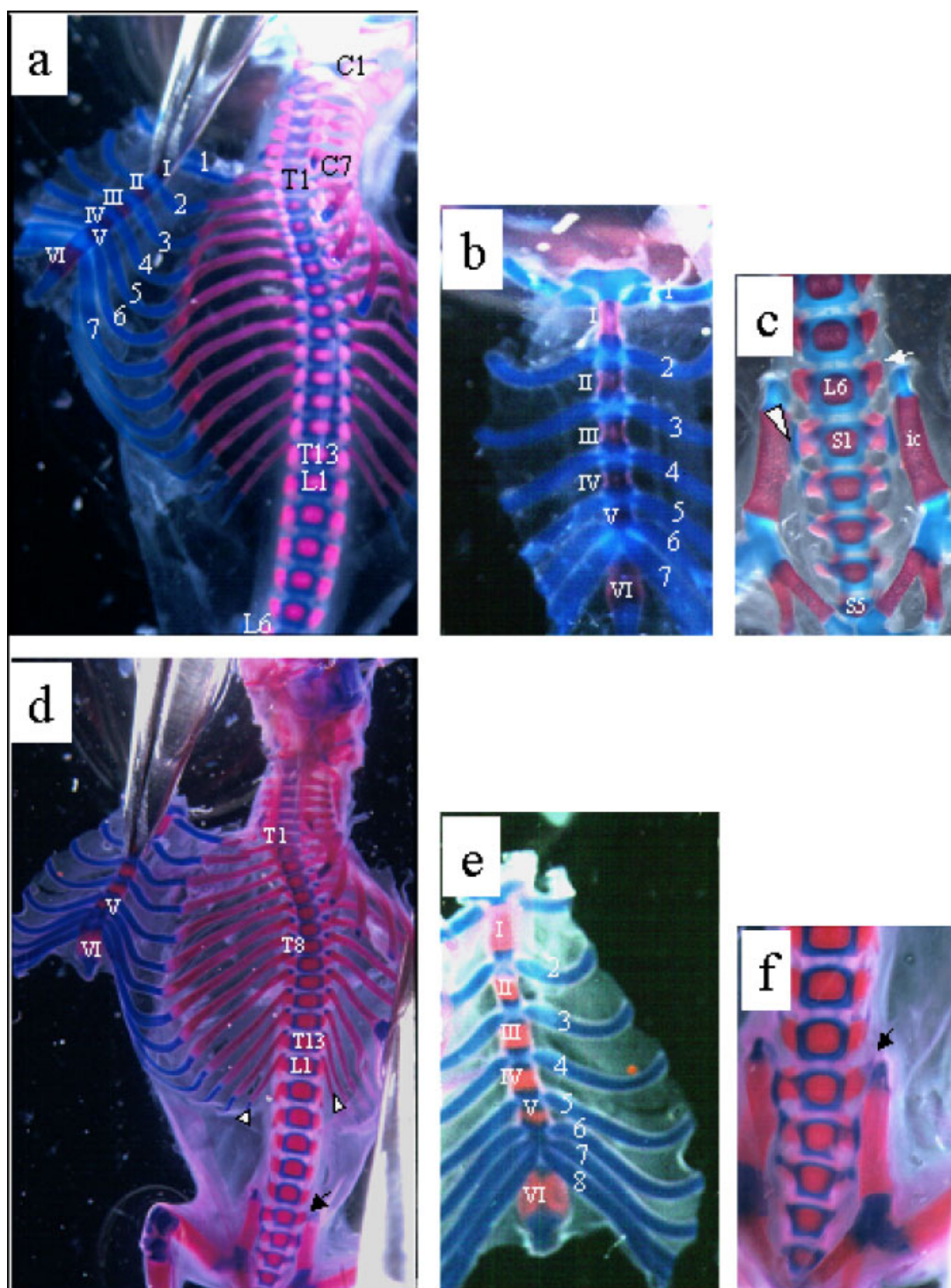


Fig. 4. a–c: Axial structure of a control fetus, as detected after the double staining method for bone (red) and cartilage (blue). a: Dorsal view of the cervical, thoracic and lumbar regions, characterized respectively by 7 (C1–C7), 13 (T1–T13), and 6 (L1–L6) vertebrae. Thoracic vertebrae articulate ribs, seven (1–7) reaching the sternum and inducing the formation of six sternebrae (I–VI) ($8\times$). b: Isolated sternum of a control fetus. Note the rib insertion and the structure of the sternebrae ($12.5\times$). c: Ventral view of the sacral region, characterized by five vertebrae (S1–S5) reaching the iliac crest (ic). Note the characteristic shape of the sacral transverse processes (arrow) in comparison to the lumbar ones (arrowhead) ($16\times$). d–f: Morphological appearance of the axial structures in a FON group fetus showing anterior homeotic transformation. A total of 13 thoracic segments are visible ($8\times$; d), the eighth showing the morphology of the seventh (the rib reaching the sternum between sternebrae V and VI, as is also well visible in e; $12.5\times$). Note the lumbar rib (arrowhead; d) and the partial homeotic anterior transformation of the first sacral vertebra, showing the left transverse process with the typical morphology of the lumbar ones (arrow), well visible in f ($16\times$).

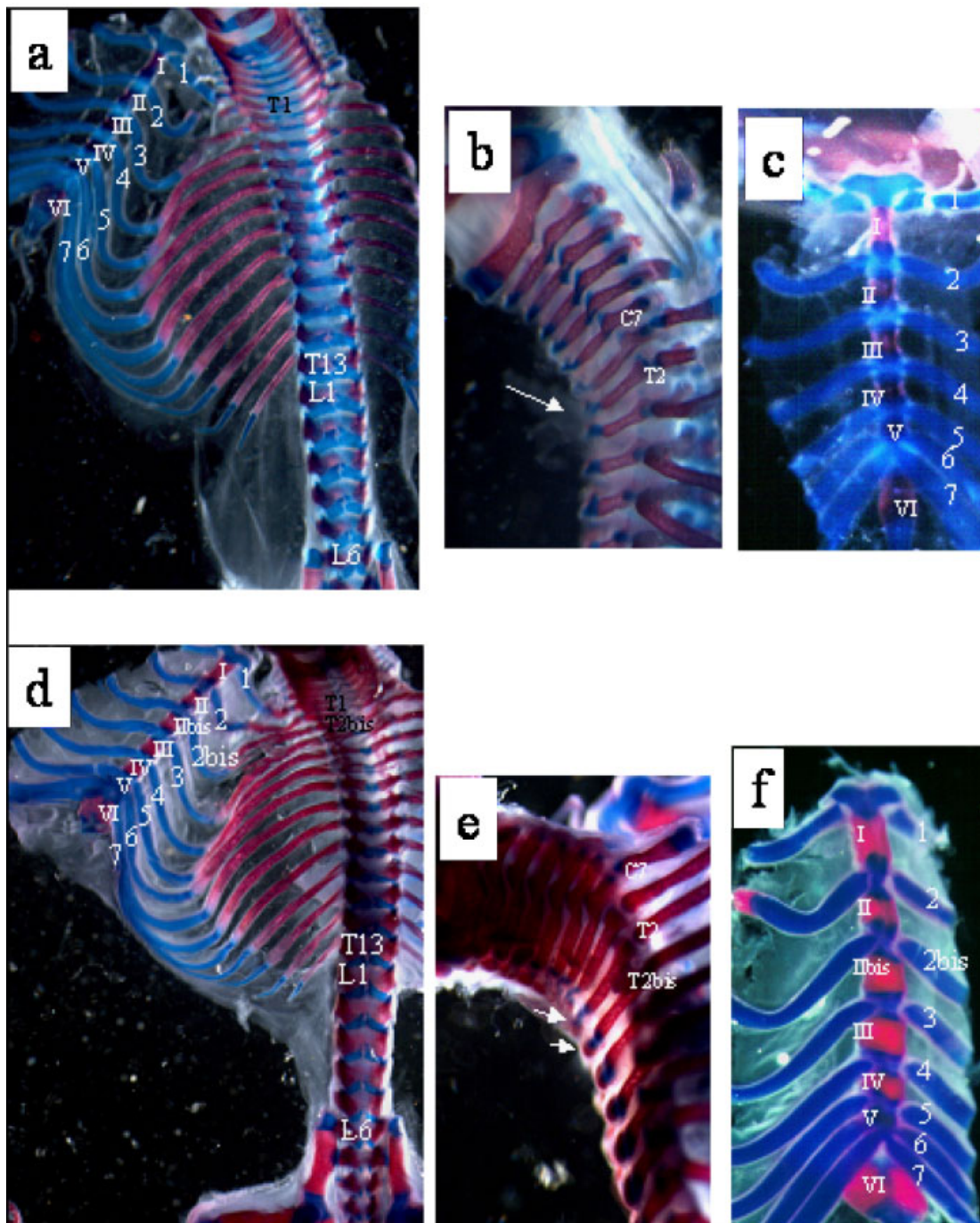


Fig. 5. a–c: Axial structure of a control fetus, as detected after the double-staining method for bone (red) and cartilage (blue). a: Dorsal view of the cervical, thoracic, and lumbar regions, characterized by 7 (C1–C7), 13 (T1–T13), and 6 (L1–L6) vertebrae, respectively. Thoracic vertebrae articulate ribs, seven (1–7) reaching the sternum and inducing the formation of six sternabrae (I–VI) ($8\times$). b: Lateral view of some cervical and thoracic vertebrae. Note the typical spine characterizing the second thoracic vertebra (arrow) ($18\times$). c: Isolated sternum of a control fetus. Note the rib insertion and the structure of the sternabrae ($12.5\times$). d–f: Morphological appearance of the axial structures in a FON group fetus showing thoracic duplication. A total of 14 thoracic segments are detectable ($8\times$; d), followed by 6 normal lumbar and 5 normal sacral vertebrae. The lateral view of some thoracic vertebrae shows the duplication of the second thoracic vertebra (arrow; e) ($18\times$). At the sternal level ($12.5\times$; f) an extra sternebra (2bis) is induced.

axial segments, or anterior morphological homeotic transformations of axial segments. Similar axial alterations have been described in the literature after treatment of pregnant mice and rats to chemicals such as retinoic

acid and valproic acid (Kessel and Gruss, 1991; Kessel, 1992; Menegola et al., 1998; Faiella et al., 2000). For all these substances, the mechanism of action could be the alteration of expression of the Hox genes implied in the

axial segment determination, by different ways of action. In particular, the teratogenic potential of valproic acid seems to be related to its inhibitory activity on the histone deacetylase enzymes, the enzymes controlling the histone compaction level and, then, the gene expression (Eikel et al., 2004a, 2004b). As far as retinoic acid is concerned, its capability to activate the axial Hox code has been clearly demonstrated (Kessel and Gruss, 1991). Due to these different ways of action, as well as to their pharmacokinetic and pharmacodynamic properties, determined also by their molecular variables, the potency and teratogenic degree of these molecules is known to be different.

At this time, there is no experimental evidence on the mechanisms implied in triazole teratogenesis. However, the effects obtained after in vitro exposure of triazoles are similar to those elicited by retinoic acid exposure in vitro (Klug et al., 1989; Lee et al., 1995) and recent data reported that subteratogenic concentrations of retinoic acid (0.025 μ M) and fluconazole (62.5 μ M) act synergistically to induce the typical branchial abnormalities in rat embryos cultured in vitro (Menegola et al., 2004).

Furthermore, the capability of triazoles to inhibit Cyp 26 activity has been demonstrated in cell cultures (Schwartz et al., 1995) and the triazole fluconazole is used clinically to enhance the plasma levels of retinoic acid in patients treated with this drug. The enzymes of Cyp 26 family, in fact, catabolize retinoic acid to the inactive forms 4-oxo- and 4-hydroxy-retinoic acid (Schwartz et al., 1995; Vanier et al., 2003). The fundamental role of Cyp 26 in maintaining the physiologic concentrations of retinoic acid is demonstrated by the distribution of Cyp 26 in several embryonic organ buds, including hindbrain and branchial arches (MacLean et al., 2001; Abu-Abed et al., 2002). There is, then, a series of circumstantial evidence showing that the teratogenic effects elicited both in vitro and in vivo by triazoles may be mediated by inhibition of Cyp 26 and by the increase of the endogenous levels of retinoic acid. In particular, the craniofacial abnormalities related to alterations of neural crest cell migration and the ectopic upper jaw cartilage, as well as the axial segment homeotic transformations, are all relatable to those elicited by in vivo and in vitro retinoic acid exposure (Klug et al., 1989; Lee et al., 1995; Menegola et al., 2004; Kessel and Gruss, 1991; Kessel, 1992).

Other compounds, as the antifungal imidazole-derivatives ketoconazole and itraconazole, are potent inhibitors of Cyp enzymes (Morita et al., 1992; Ervine et al., 1996; Colburn et al., 2004). As a consequence, on the basis of results on triazoles, their teratogenic effects, mediated by endogenous retinoic acid imbalance, cannot be excluded. Furthermore, as retinoic acid acts as a morphogen in different embryonic compartments and in different developmental stages, it is likely that the administration of triazoles or other Cyp inhibitors during a number of morphogenetic processes could result in a wide spectrum of abnormalities.

In conclusion, the results of this experiment show the intrinsic teratogenic potential of FON, and demonstrate that in vitro whole-embryo culture experiments may be useful in predicting the teratogenic potential of a class of chemicals, even if they currently have a limited value for risk assessment in humans. On

the basis of the in vitro studies, the teratogenic potential of FON can be extended to the whole class of triazole derivatives. Taking into account the large diffusion of these chemicals in the environment and their use in human therapy, the safety of this class of substances for the human conceptus should be, in our opinion, reconsidered.

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